

Banking Fraud Detection & Risk Analytics System

SQL Analytics Questions

SECTION 1 — USER & ACCOUNT ANALYTICS

1. How many unique users are registered in the system?
2. Which users own multiple bank accounts?
3. Which account types have the highest average balances?
4. Which users maintain the highest total account balances?
5. How many users are currently suspended or dormant?
6. What is the distribution of account statuses across the system?

SECTION 2 — TRANSACTION ANALYTICS

1. What is the total transaction volume processed by the system?
2. Which transaction types occur most frequently?
3. Which accounts perform the highest number of transactions?
4. What are the monthly transaction trends?
5. Which transactions have exceptionally high amounts?
6. Which locations generate the highest transaction activity?

SECTION 3 — FRAUD & RISK ANALYTICS

1. Which users have the highest fraud risk scores?
2. Which transactions have the highest fraud risk?
3. Which countries generate the most fraud alerts?
4. Which devices are linked to suspicious transactions?

5. What percentage of transactions are marked as suspicious?
6. Which users repeatedly appear in fraud alerts?

SECTION 4 — SECURITY & BEHAVIORAL ANALYTICS

1. Which users have the highest number of failed login attempts?
2. Which devices are associated with suspicious login behavior?
3. What are the most common login failure reasons?
4. Which users access the system from multiple locations?
5. Which devices are used by multiple users?
6. What are the peak login activity hours?

SECTION 5 — ADVANCED SQL & DATABASE FEATURES

1. Rank users based on total transaction amounts using window functions.
2. Identify monthly transaction growth using the LAG() window function.
3. Detect top risky users using CTE-based analytics.
4. Create a view for high-risk users and suspicious transactions.
5. Implement a stored procedure for fraud detection and risk analysis.
6. Create a trigger to automatically generate fraud alerts for high-risk transactions.

BUSINESS INSIGHTS

1. High-value transactions contribute disproportionately to fraud alerts.
2. Users with repeated failed login attempts demonstrate elevated fraud risk.
3. Devices shared across multiple users indicate suspicious behavioral patterns.
4. Foreign transaction locations showed higher average fraud risk scores.
5. Dormant and suspended users still demonstrated transactional activity.
6. Critical-risk users perform significantly larger transaction volumes.